Spring 2026 Syllabus

CDA 4621: Control of Mobile Robots (Undergraduate, 3 Credits)
CAI 5815: Autonomous Mobile Robots (Graduate, 3 Credits)

Prof. Alfredo Weitzenfeld

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Course Prerequisites: COP 3514 (min grade B) (undergrads only)

Class Schedule: Tue & Thu 3:30pm-4:45pm, CHE 303

Prof. Office Hours: Tue & Thu 2:00pm-3:15pm, Biorobotics Lab, ENG 122B

TA: Chance Hamilton**, Email:** chamilton4@usf.edu, **Office Hours:** Thu 10:30am-1pm, ENG 122B

TA: Brendon Johnson**, Email:** brendon45@usf.edu, **Office Hours:** Wed 2pm-4pm, ENG 122B

Biorobotics Lab: ENG 122B

Student Physical Robot Lab: ENG 202

UNIVERSITY COURSE DESCRIPTION (USF Catalog)

Mobile Robotic Control Systems design and implementation. Includes microcontroller, sensor, and actuator control processes for localization and navigation.

COURSE OBJECTIVES

The course objectives are to learn and apply the underlying principles of the control of autonomous mobile robots.

STUDENT LEARNING OUTCOMES

Students will learn the theory and practice of autonomous mobile robots, including kinematics, motion to goal, wall following, obstacle avoidance, PID, localization, particle and Kalman filters, mapping, path planning, and reinforcement learning (RL), and other selected topics.

FIRST DAY ATTENDANCE (FDA)

Attend class Tue Jan 13 and take Syllabus Quiz by Tue Jan 13 11:59pm. Students need to get a perfect score on the quiz to avoid being dropped from the class. Students may take the quiz multiple times.

ATTENDANCE POLICY

Regular class attendance is not required, except for first day of classes and exams.

WORKLOAD

Make sure you have time for the class, as programming labs cannot be completed at the last moment.

TEXTBOOK

There is no textbook for the course. All material covered in class is posted in Canvas.

PROGRAMMING LANGUAGE: Python

ROBOTIC INFRASTRUCTURE

1. **"Webots" Robot Simulator** (individual students): "Webots (R2025a)" (www.cyberbotics.com – free) + "FAIRIS" (https://github.com/biorobaw/FAIRIS-Lite/)
2. **Physical Robot** (one or two students per group): Raspberry Pi Robot (https://github.com/biorobaw/HamBot)

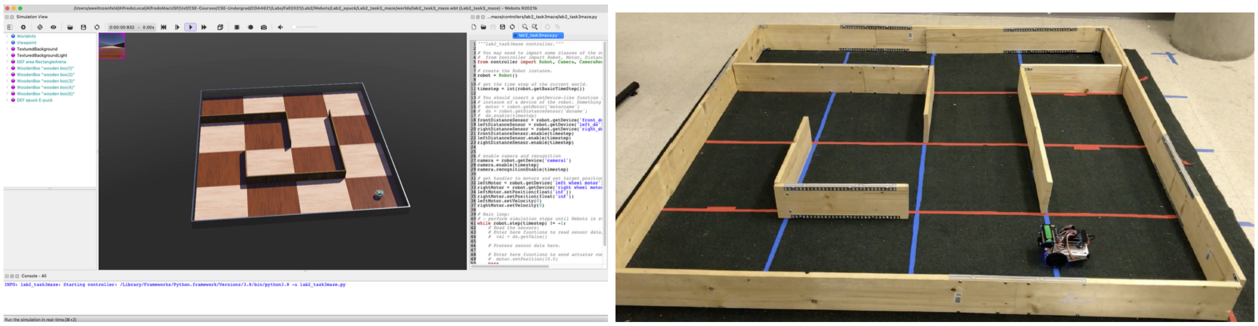

Figure 1. (Left) "Webots" Robot Simulator, (Right) Physical Robot.

PHYSICAL ROBOTS

Work with physical robots is optional, based on availability, and in groups of one or two students. Students interested in using a physical robot need to complete and submit the following forms and quiz:

- Forms: (a) RobotLoan and (2) Lithium-Ion Batteries
- Quiz: Robot Lithium-Ion Batteries (You may take it multiple times until getting a perfect score.)

Students that do not return the robot by the end of the semester will get an "F" as final grade. Note that students may switch back to Webots at any time.

GRADE COMPUTATION

Grading is based on labs, exams, and quizzes, for a total of **110%**.

- **Labs (60%) – Each lab (top 4 out of 5) is worth 15%**

Labs	Lab Due Date	Material
Lab 1	2/9	Kinematics
Lab 2	3/2	PID Navigation Control
Lab 3	3/30	Motion Planning
Lab 4	4/20	Localization & Filters
Lab 5	5/4	Mapping & Path Planning

- **Theory Exams (40%) – Each exam is worth 20%**

Theory Exams	Exam Date	Material
Midterm Exam	2/12	Non-Probabilistic Robotics (Ch 5-7)
Final Exam	5/7	Probabilistic Robotics (Ch 8-14)

- **Quizzes (10%)**

Quizzes	Dates	Material
In-Class Quizzes	Varies	All

- **Project – Grad Student Only (0%)**

Graduate students are required to submit and present in class an individual robotics project. If not submitted or presented, 10 points will be deducted from final grade. Grad projects questions will be included in the final exam.

COMMUNICATION

Make sure you continuously monitor announcements and emails. Preferred direct communication with the Prof and TAs is via email, usually answered within 24 hours, except weekends.

LAPTOP COMPUTER REQUIREMENT

It is a requirement for this class to bring a laptop to class for all exams, with working camera and microphone if using Honorlock. Note that Webots and physical robots need use of a laptop.

HONORLOCK

Exams may require use of Honorlock. Be prepared to have computer with a webcam and a microphone. You should not take this class if you do not agree to using Honorlock.

GRADING POLICIES

- **Exam Requirements.** All exams must be taken in person in class during regular scheduled class hours. You cannot take exams at a different schedule or location. You cannot take exams remotely. If you miss any exam for any reason, you will receive a "0" for the exam. You need to bring a laptop to class for all exams. If you do not own a laptop, you need to arrange for one, no exceptions.
- **Lab Submissions**. Labs must be uploaded to Canvas by the due date and time. There is a grade penalty of 20% per day for late lab submissions, independent of reason. After 5 days, lab grade becomes "0". Partial submission will receive a "0" for any missing sections that are uploaded after the lab due date. It is the student responsibility to check for completeness and upload correct files. No handwritten text or diagrams will be accepted. There will be no extensions to lab due dates.
- **Lab Presentations**. In addition to lab submissions, students are required to schedule a Q&A presentation with the TA. Students who do not attend the required presentation on the specified date will get a "0" on the full lab.
- **Top 4 out of 5 Lab Grades.** Lab grades will be based on top 4 out of 5 lab grades. Lowest lab grade will be dropped from total lab grade computation. (Note that plagiarism in any of the labs will result in forfeiting the "top 4 out of 5 lab grades" option.)
- **Individual Work**. All labs are individual, except for students in the same group working with physical robots. All students must submit original work. Submissions will be checked for plagiarism.
- **Quizzes**. There will be in-class quizzes to be held at the end of classes and may vary in number. To take the quiz you will need to bring your laptop. You are not allowed to take quizzes remotely or outside class schedule. Final quiz grade will be based on the sum of all obtained quiz points.
- **Extra Credit.** Quizzes count as extra credit. No additional extra credit is offered in the course.
- **No Makeups**. There are no make-ups for exams, labs, or quizzes. If you miss any of them, you will get a "0" for them, independent of reason.
- **No Excuses**. No excuses will be accepted for missing exams, labs, or quizzes. You are responsible for attending and submitting everything in time.
- **No "Incomplete" Grades**. No "Incomplete" grades are available for the class. No material will be graded after semester ends. A final grade will be submitted immediately at the end of the semester.

- **Grade Grievances**. There is a 1-week time limit for reviewing lab or exam grades after they are posted. No grades will be discussed after a week of their posting dates.
- **No Grade Rounding**. No points will be added at the end of the semester to go up from one letter grade to another. For example, if you receive 79.99, your final grade will be C+. Grades will not be modified based on how close you are to the next letter grade or what your GPA is. Do not send emails to Prof or TA requesting grade changes.
- **Use of AI Tools:** If using AI tools, adherence to the following is required:
  1. **Code Segments**: Indicate in the code file all AI-generated code segments.
  2. **Lab Quizzes**: Lab submissions include an accompanying quiz that needs to be completed.
- **Final Grade Scale**: Final course grades will be based on the following scale:

Grade Points	Letter Grade
92.50-100	A
90.00-92.49	A-
87.50-89.99	B+
82.50-87.49	B

Grade Points	Letter Grade
80.00-82.49	B-
77.50-79.99	C+
72.50-77.49	C
70.00-72.49	C-

Grade Points	Letter Grade
67.50-69.99	D+
62.50-67.49	D
60.00-62.49	D-
0-59.99	F

ACADEMIC INTEGRITY

Academic integrity is taken very seriously. See USF regulations and Bellini College specific policies:
https://www.usf.edu/undergrad/students/ethics-integrity.aspx
https://www.usf.edu/graduate-studies/students/academic-integrity-of-students/

- **Academic Integrity Grading for Exams:** Any level of cheating in an exam will result in an automatic "F" for the exam and "F" and possibly a "FF" for the course.
- **Academic Integrity Grading for Labs:** All labs must be original and developed by the student. Unacknowledged use of external code or resources, including use of AI tools like ChatGPT or CoPilot, is considered plagiarism, unless clear and precise references are provided with links, prompts, and results that can be verified. If generating code with AI tools exact prompts and results must be submitted. <u>It will be considered plagiarism if more than one student submits the same code independent of how it was generated</u>, including use of AI tools with the same prompt, output, or any code from the same source. Copying existing code from the Web, such as Github, is also considered plagiarism. <u>Sharing or posting your lab code or solutions for others to copy are considered academic dishonesty and will receive same penalty as plagiarism</u>. Lab grades will be based on actual code developed by the student. Any level of plagiarism in the labs will result in an automatic "F" for the lab and an "F" and possibly "FF" for the course. <u>Note that it is better to get a partial grade on your own work than risk an "F" and possibly "FF" for the course due to plagiarized work.</u>
- **Lab Grade Deductions:** The table below reflects the lab and course final grade deductions based on Moss similarity reporting between multiple student submissions or unreported external sources (note that plagiarism checks will not be limited to Moss and the gravity and level of infraction depend on the Professor's assessment and may result in an automatic "F" or possibly "FF" for the course):

Lab Moss Report (Per Task)	Lab Grade	Course Grade Deduction
50% < similarity < 74%	0	10%
75% < similarity < 100%	0	20%

Final course grade may become an "F" and even an "FF" based on number and level of infractions.

CLASS RECORDINGS

Professor does not record now allow classes to be recorded. Note that per USF Regulation 10.048 you are required to obtain the Prof permission for any recordings. If allowed by the Prof, class recordings are for your own personal use, and cannot to be shared, fully or partially, in any media or form.

USF POLICY STATEMENT ON HONORLOCK

All students must review the syllabus and the requirements, including the online terms and video testing requirements, to determine if they wish to remain in the course. Enrollment in the course is an agreement to abide by and accept all terms. Any student may elect to drop or withdraw from this course before the end of the drop/add period. Online exams and quizzes within this course may require online proctoring. Therefore, students will be required to have a webcam (USB or internal) with a microphone when taking an exam or quiz. Students understand that this remote recording device is purchased and controlled by the student and that recordings from any private residence must be done with the permission of any person residing in the residence. To avoid any concerns in this regard, students should select private spaces for the testing. The University library and other academic sites at the University offer secure private settings for recordings and students with concerns may discuss location of an appropriate space for the recordings with their instructor or advisor. Students must ensure that any recordings do not invade any third-party privacy rights and accept all responsibility and liability for violations of any third-party privacy concerns. Students are strictly responsible for ensuring that they take all exams using a reliable computer and high-speed internet connection. Setup information will be provided prior to taking the proctored exam. To use Honorlock, students are required to download and install the Honorlock Google Chrome extension. For additional information please visit the USF online proctoring student FAQ and Honorlock student resources.

COUNCELING CENTER

The USF Counseling Center offers clinical services that are free, confidential, and available to all currently registered USF undergraduate and graduate students.

USF STANDARD PROCEDURES

Policies about disability access, religious observances, academic grievances, academic integrity and misconduct, academic continuity, food insecurity, sexual harassment, and Covid-19 are governed by a central set of policies that apply to all classes at USF. All students must comply with university policies and posted signs regarding COVID-19 mitigation measures, including wearing face coverings and maintaining social distancing during in-person classes. Failure to do so may result in dismissal from class, referral to the Office of Student Conduct and Ethical Development, and possible removal from campus. These policies can be accessed at: https://www.usf.edu/provost/faculty-success/resources-policies-forms/core-syllabus-policy-statements.aspx

USF ADDITIONAL PROCEDURES

- Students with disabilities should see SAS to obtain appropriate accommodations.
- Disruption of Academic Process: USF Regulation 3.025
- Student Code of Conduct: USF Regulation 6.0021

CLASS SCHEDULE (Subject to Change)

Week	Topic	Lab [Due Date]	Canvas Chapter	Date
1	Intro to Course & Mobile Robotics		1, 2	1/13
	Robot Hardware & Webots		3	1/15
2	Webots-FAIRIS	Lab1 [2/9]	3	1/20 (TA)
	Kinematics		5	1/22
3	Kinematics		5	1/27
	Kinematics		5	1/29
4	Kinematics		5	2/3
	PID		6	2/5
5	Wall Following	Lab2 [3/2]	6	2/10 (TA)
	Midterm Exam			2/12
6	Basic Bug Algorithms		7	2/17
	Tangent Bug Algorithms		7	2/19
7	Tangent Bug Algorithms		7	2/24
	Intro Probabilistic Robotics		8	2/26
8	Wall Localization	Lab3 [3/30]	9	3/3
	Landmark Localization		9	3/5
9	Motion Planning (Motion-To-Goal)		7	3/10 (TA)
	Motion Planning (Obstacle Avoidance)		7	3/12 (TA)
	Spring Break			*3/16 - 3/20*
10	Particle Filter		10	3/24
	Particle Filter		10	3/26
11	Kalman Filter	Lab4 [4/20]	11	3/31
	Kalman Filter		11	4/2
12	Mapping		12	4/7
	Mapping		12	4/9
13	Path Planning		13	4/14
	Path Planning		13	4/16
14	Reinforcement Learning	Lab5 [5/4]	14	4/21
	Reinforcement Learning		14	4/23
15	Grad Student Projects			4/28
	Grad Student Projects			4/30
	Final Exam (12:30-2:30pm)			5/7